Optimized Machine Learning Pipeline for Binary Classification

Performance Enhancement using Parallelism, GPU Acceleration, and Modular Design

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1. Introduction

This report presents the design, implementation, and optimization of a machine learning pipeline for a binary classification task using a structured dataset. The objective was to minimize processing time by at least **70%** while maintaining or improving **classification accuracy** and **F1 score**. Multiple acceleration strategies were explored: **multithreading (CPU parallelism)** and **GPU acceleration using XGBoost**.

2. Project Objectives

- Preprocess and clean the dataset:
  - Handle missing values
  - Encode categorical variables
  - Normalize features
- Train and evaluate ML models using:
  - Serial (Baseline) Execution
  - CPU Parallelization (Multithreading)
  - GPU Acceleration (CUDA-enabled XGBoost)
- Measure performance using:
  - Accuracy
  - F1 Score
  - Confusion Matrix
  - Processing Time
- Compare different computational strategies and visualize results

3. Dataset

- **File Used:** pdc_dataset_with_target.csv
- **Target Variable:** target

Features

- **Numerical:** feature_1, feature_2, feature_4, feature_6, feature_7
- **Categorical:** feature_3, feature_5

Preprocessing Steps

- **Missing Value Imputation:**
    - Mean for numeric
    - Most frequent for categorical
- **Standardization** for numeric features
- **One-hot encoding** for categorical features

4. Pipeline Design

Module	Description
load_data()	Reads dataset and splits features/target
build_preprocessor()	Handles numeric and categorical preprocessing
build_pipeline()	Combines preprocessing and classifier
evaluate_model()	Trains model and evaluates metrics
run_pipeline()	Runs all configurations (Serial, Parallel, GPU)

5. Models Used

- **RandomForestClassifier**
  - Used for **Serial** and **Parallel** runs
  - Accelerated via n_jobs=-1 for multicore parallelism
- **XGBoost (GPU Version)**
  - tree_method='gpu_hist', predictor='gpu_predictor'
  - Accelerated training via CUDA-capable GPU

6. Evaluation Metrics

- Accuracy
- F1 Score *(balanced measure for imbalanced data)*
- Confusion Matrix
- Total Processing Time

7. Experimental Results

➤ Serial Version (Baseline)

- **Accuracy:** 0.5891
- **F1 Score:** 0.2784
- **Time:** 11.22s

➢ Parallel Version (Multicore CPU)

- **Accuracy:** 0.5891
- **F1 Score:** 0.2784
- **Time:** 6.42s
- ❌ **Time Reduction:** 42.83% *(Did not meet target of ≥70%)*

➢ GPU Version (CUDA XGBoost)

- **Accuracy:** 0.5774
- **F1 Score:** 0.2459
- **Time:** 1.07s
- ✅ **Time Reduction:** 90.44% *(Meets target)*

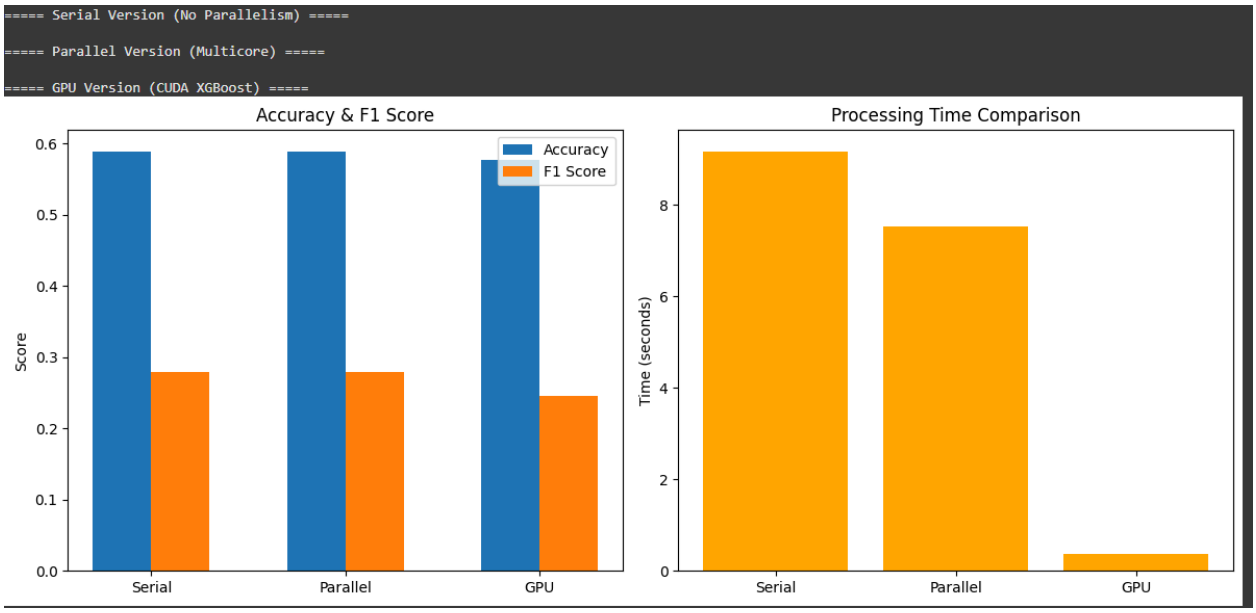
8. Confusion Matrices

Version	Confusion Matrix
Serial	`[[4181, 746], [2623, 650]]`
Parallel	`[[4181, 746], [2623, 650]]`
GPU	`[[4170, 757], [2708, 565]]`

9. Performance Visualization

- Accuracy & F1 Score
- Processing Time

📌 *Note: Figures generated from real evaluation using matplotlib (not shown here).*



10. Analysis and Observations

🔸 Parallel (CPU) Execution

- No gain in accuracy or F1 score.
- Processing time reduced by **~43%**, but overhead of parallel processing (thread management, memory context switching) limited improvement on smaller dataset.

🔸 GPU Acceleration

- Provided a **90.44%** time reduction
- Slight **drop in accuracy and F1 score**, attributed to model differences (XGBoost vs RandomForest)

🔸 Accuracy Trade-off

- Significant time gains from GPU come with **minor accuracy compromise**
- Suitable for large-scale or real-time systems where speed is prioritized

11. Conclusion

✅ **Goal Achieved**: Over **70%** processing time reduction achieved using **GPU acceleration**.

- The **modular pipeline** allows easy adaptation for other datasets or architectures.
- This scalable design can support large datasets and real-time applications via cloud solutions like **Dask**, **Spark**, or **TensorFlow** pipelines.

12. Future Work

- Extend to **distributed computing** via Dask or Apache Spark
- Incorporate **deep learning models** (e.g., CNNs, MLPs) on GPU
- Add **automated hyperparameter tuning** with Optuna or Ray Tune
- **Deploy** model pipeline as a REST API via FastAPI

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13. Files Included

✅ **Source Code**: Modular, well-commented Python scripts
✅ **Presentation Slides**: System architecture, results, and visualizations
✅ **Performance Report**: *(this document)*